

## LAPPEENRANTA KONNUNSUO, Finland

WMO: 027330

Lat: 61.039N

Lon: 28.564E

Elev: 152

StdP: 14.62

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -21.2      | -14.6 | -27.1                         | 1.2 | -21.1 | -20.6 | 1.8 | -13.1 | 22.6                  | 30.2 | 20.7 | 28.6 | 1.2                   | 320  | 0.648 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.7                   | 81.7            | 66.7 | 78.4 | 65.0 | 75.0 | 62.8 | 69.5                | 78.1 | 67.2 | 75.5 | 64.7 | 72.3 | 7.7                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.2                            | 97.1 | 73.5 | 63.8 | 89.2 | 72.3 | 61.4 | 81.7 | 69.4 | 33.5          | 78.4 | 31.7 | 76.0 | 29.8 | 72.0 | 76.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 20.3              | 18.1 | 16.2 | -24.8                      | 85.0 | 8.8                | 4.9 | -31.1                                              | 88.6 | -36.3      | 91.5 | -41.2      | 94.2 | -47.5      | 97.8 |  |  |
|                   |      |      | -26.0                      | 71.9 | 8.5                | 3.0 | -32.1                                              | 74.1 | -37.1      | 75.8 | -41.8      | 77.5 | -48.0      | 79.7 |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 39.2   | 15.2  | 19.0  | 26.0  | 37.3  | 49.6  | 56.3  | 61.4  | 58.1  | 50.2  | 38.7  | 32.5  | 24.5  |      |
|                                                                                  | DBStd                                                                            | 17.81  | 15.09 | 13.75 | 9.62  | 6.81  | 7.95  | 6.38  | 5.97  | 5.59  | 6.00  | 6.78  | 7.88  | 11.73 |      |
|                                                                                  | HDD50                                                                            | 4945   | 1080  | 869   | 745   | 383   | 105   | 16    | 1     | 5     | 69    | 354   | 526   | 792   |      |
|                                                                                  | HDD65                                                                            | 9482   | 1545  | 1289  | 1210  | 832   | 479   | 268   | 145   | 223   | 443   | 815   | 976   | 1257  |      |
|                                                                                  | CDD50                                                                            | 993    | 0     | 0     | 0     | 1     | 93    | 206   | 356   | 257   | 76    | 4     | 0     | 0     |      |
|                                                                                  | CDD65                                                                            | 55     | 0     | 0     | 0     | 0     | 2     | 8     | 35    | 10    | 0     | 0     | 0     | 0     |      |
|                                                                                  | CDH74                                                                            | 916    | 0     | 0     | 0     | 0     | 73    | 185   | 474   | 183   | 1     | 0     | 0     | 0     |      |
|                                                                                  | CDH80                                                                            | 167    | 0     | 0     | 0     | 0     | 8     | 27    | 100   | 32    | 0     | 0     | 0     | 0     |      |
| Wind                                                                             | WSAvg                                                                            | 7.5    | 7.7   | 7.7   | 8.3   | 7.3   | 7.9   | 7.6   | 6.2   | 6.1   | 6.6   | 7.5   | 8.1   | 8.7   |      |
| Precipitation                                                                    | PrecAvg                                                                          | 25.4   | 1.9   | 1.5   | 1.4   | 1.2   | 1.7   | 2.3   | 2.8   | 2.9   | 2.4   | 2.6   | 2.5   | 2.2   |      |
|                                                                                  | PrecMax                                                                          | 31.3   | 3.1   | 4.0   | 2.6   | 2.6   | 5.6   | 3.9   | 6.4   | 5.8   | 5.5   | 5.8   | 5.2   | 5.1   |      |
|                                                                                  | PrecMin                                                                          | 19.8   | 0.4   | 0.2   | 0.2   | 0.4   | 0.2   | 0.7   | 0.6   | 0.4   | 0.9   | 0.6   | 0.4   | 0.8   |      |
|                                                                                  | PrecStd                                                                          | 3.1    | 0.7   | 0.8   | 0.7   | 0.6   | 1.2   | 0.8   | 1.3   | 1.5   | 1.4   | 1.2   | 1.0   | 0.9   |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 37.5  | 40.4  | 48.4  | 65.3  | 80.6  | 83.6  | 87.0  | 83.8  | 70.6  | 57.1  | 50.5  | 46.5  |      |
|                                                                                  |                                                                                  | MCWB   | 35.5  | 36.4  | 38.3  | 50.9  | 60.8  | 65.4  | 70.1  | 69.3  | 59.3  | 52.3  | 47.1  | 44.3  |      |
|                                                                                  | 2%                                                                               | DB     | 35.9  | 37.2  | 43.0  | 59.0  | 75.4  | 79.1  | 82.7  | 79.0  | 67.4  | 54.0  | 46.8  | 41.5  |      |
|                                                                                  |                                                                                  | MCWB   | 34.3  | 35.1  | 36.7  | 47.0  | 58.6  | 63.9  | 68.8  | 65.4  | 58.2  | 49.9  | 44.5  | 39.1  |      |
|                                                                                  | 5%                                                                               | DB     | 34.0  | 35.3  | 39.5  | 54.1  | 71.1  | 75.2  | 79.4  | 75.0  | 64.3  | 52.0  | 44.4  | 38.9  |      |
|                                                                                  |                                                                                  | MCWB   | 32.4  | 33.5  | 34.9  | 44.4  | 56.1  | 61.6  | 67.4  | 63.1  | 56.7  | 48.6  | 42.7  | 36.9  |      |
|                                                                                  | 10%                                                                              | DB     | 32.7  | 33.9  | 37.1  | 49.6  | 67.0  | 71.4  | 75.6  | 71.3  | 61.7  | 49.9  | 42.1  | 36.4  |      |
|                                                                                  |                                                                                  | MCWB   | 31.2  | 32.5  | 33.9  | 42.2  | 54.1  | 60.3  | 65.3  | 61.7  | 55.8  | 46.8  | 40.5  | 34.5  |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 36.0  | 36.8  | 39.9  | 51.7  | 63.5  | 69.6  | 73.3  | 71.0  | 62.4  | 53.7  | 47.3  | 44.5 |
|                                                                                  |                                                                                  |        | MCDB  | 37.1  | 38.7  | 45.5  | 63.4  | 75.3  | 78.7  | 82.1  | 81.4  | 68.2  | 56.4  | 49.9  | 47.0 |
| 2%                                                                               |                                                                                  | WB     | 34.7  | 35.2  | 37.5  | 48.7  | 60.5  | 66.2  | 70.6  | 67.2  | 60.1  | 51.0  | 44.7  | 39.6  |      |
|                                                                                  |                                                                                  | MCDB   | 35.8  | 37.1  | 41.4  | 56.4  | 70.7  | 76.4  | 79.7  | 75.7  | 64.4  | 53.3  | 46.6  | 41.4  |      |
| 5%                                                                               |                                                                                  | WB     | 33.0  | 33.9  | 35.9  | 45.5  | 58.0  | 63.1  | 68.6  | 64.8  | 58.1  | 49.0  | 42.9  | 37.0  |      |
|                                                                                  |                                                                                  | MCDB   | 34.0  | 35.2  | 38.8  | 53.0  | 68.6  | 72.5  | 76.9  | 72.3  | 62.2  | 51.6  | 44.5  | 38.8  |      |
| 10%                                                                              |                                                                                  | WB     | 31.6  | 32.6  | 34.4  | 43.1  | 55.4  | 61.0  | 66.6  | 62.6  | 56.6  | 46.9  | 40.8  | 34.9  |      |
|                                                                                  |                                                                                  | MCDB   | 32.8  | 33.8  | 36.8  | 49.0  | 65.1  | 69.9  | 74.3  | 69.9  | 60.7  | 49.8  | 42.1  | 36.3  |      |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB                                                                            | MDBR   | 13.2  | 13.1  | 16.1  | 18.8  | 22.7  | 21.0  | 20.7  | 21.8  | 17.8  | 12.6  | 7.6   | 8.7   |      |
|                                                                                  |                                                                                  | MCDBR  | 6.6   | 9.3   | 16.8  | 27.7  | 31.7  | 28.6  | 27.3  | 29.1  | 23.1  | 16.7  | 9.8   | 8.3   |      |
|                                                                                  |                                                                                  | MCWBR  | 6.7   | 8.3   | 11.9  | 17.1  | 17.1  | 15.8  | 15.2  | 17.1  | 15.2  | 13.2  | 9.3   | 7.9   |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 6.4   | 9.0   | 13.6  | 24.0  | 27.0  | 25.1  | 24.8  | 25.7  | 18.8  | 13.7  | 9.4   | 7.9   |      |
|                                                                                  |                                                                                  | MCWBR  | 6.5   | 8.2   | 10.2  | 15.7  | 15.9  | 14.8  | 14.5  | 15.8  | 13.1  | 11.2  | 8.9   | 7.7   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.259  | 0.301 | 0.326 | 0.365 | 0.361 | 0.358 | 0.383 | 0.370 | 0.345 | 0.327 | 0.284 | 0.236 |       |      |
|                                                                                  | taud                                                                             | 2.070  | 2.121 | 2.193 | 2.325 | 2.423 | 2.461 | 2.409 | 2.434 | 2.494 | 2.394 | 2.162 | 2.014 |       |      |
|                                                                                  | Ebn at Noon                                                                      | 143    | 209   | 248   | 262   | 274   | 276   | 265   | 259   | 246   | 205   | 134   | 98    |       |      |
|                                                                                  | Edh at Noon                                                                      | 13     | 23    | 30    | 33    | 32    | 32    | 33    | 30    | 23    | 18    | 12    | 8     |       |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 77     | 257   | 718   | 1171  | 1609  | 1691  | 1653  | 1274  | 745   | 299   | 81    | 40    |       |      |
|                                                                                  | RadStd                                                                           | 14     | 60    | 97    | 127   | 187   | 112   | 154   | 147   | 81    | 42    | 13    | 5     |       |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (33 neighbors) | +0.94 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | -333        | -320  | N/A   | N/A   |

Nomenclature: See separate page